

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/01

Paper 1 Multiple Choice

15 September 2017

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

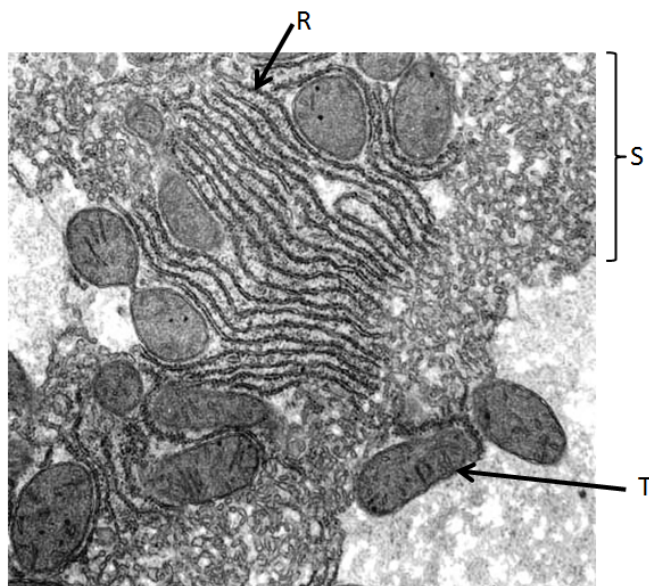
This document consists of **23** printed pages and **1** blank page.



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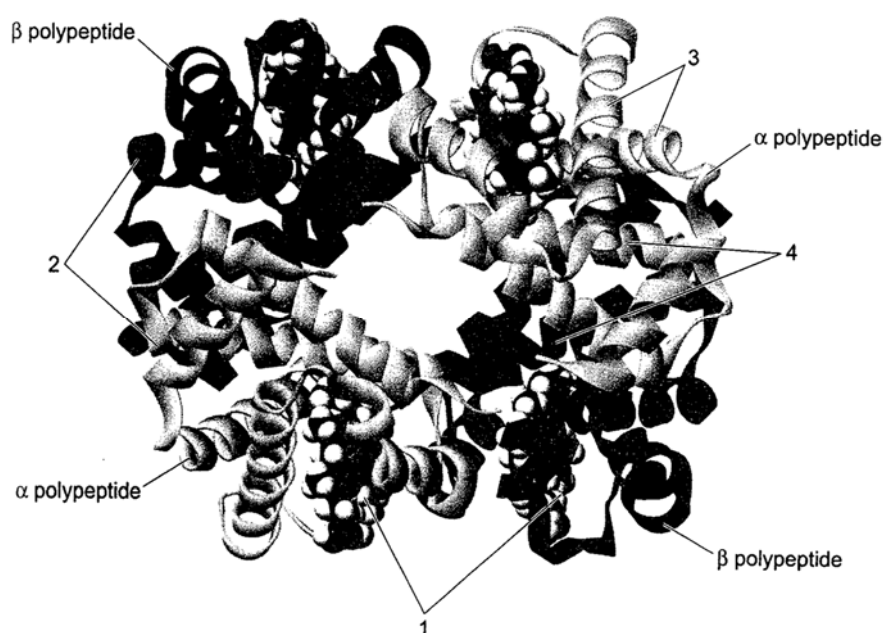
- 1 The figure below shows an electron micrograph of an eukaryotic cell.

Which of the following option correctly matches the structures **R**, **S** and **T** to their respective functions?



	R	S	T
A	Involved in proteins glycosylation	Site of lipid synthesis	To convert light energy to chemical energy
B	Site of protein synthesis	Site of detoxification reaction	Supplying cellular energy
C	Site of detoxification reaction	Involved in protein glycosylation	Remove worn out organelles
D	Site of protein synthesis	Contains proteins to be secreted	Supplying cellular energy

2 The diagram shows a haemoglobin molecule.



Which identifies the different parts of the molecule?

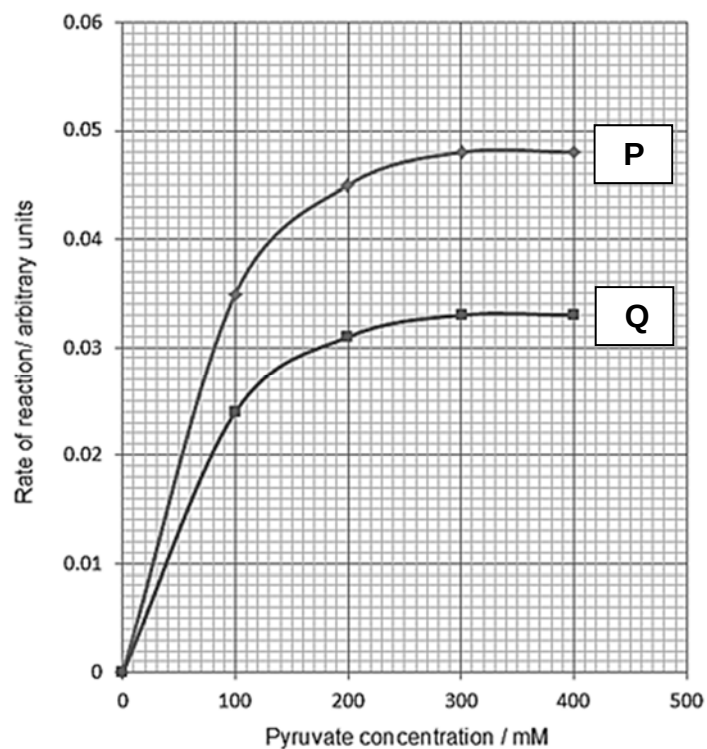
	1	2	3	4
A	prosthetic group	beta pleated sheet	alpha helix	hydrophilic amino acid
B	hydrophobic amino acids	beta pleated sheet	prosthetic group	binding site
C	prosthetic group	hydrophilic amino acids	alpha helix	hydrophobic amino acids
D	prosthetic group	hydrophilic amino acid	binding site	hydrophobic amino acids

3 Which of the following statements about enzymes are false?

- 1 All enzymes are globular proteins.
- 2 Enzymes catalyse reactions by decreasing the activation energy.
- 3 A prosthetic group is tightly bound to the enzyme, while a coenzyme is a loosely bound to the enzyme.
- 4 The effect of competitive inhibitors can be reduced by increasing substrate concentration.
- 5 An allosteric binding site refers to the active site that has undergone an induced fit.

- A** 1 and 3 only
B 1 and 5 only
C 2 and 4 only
D 3 and 5 only

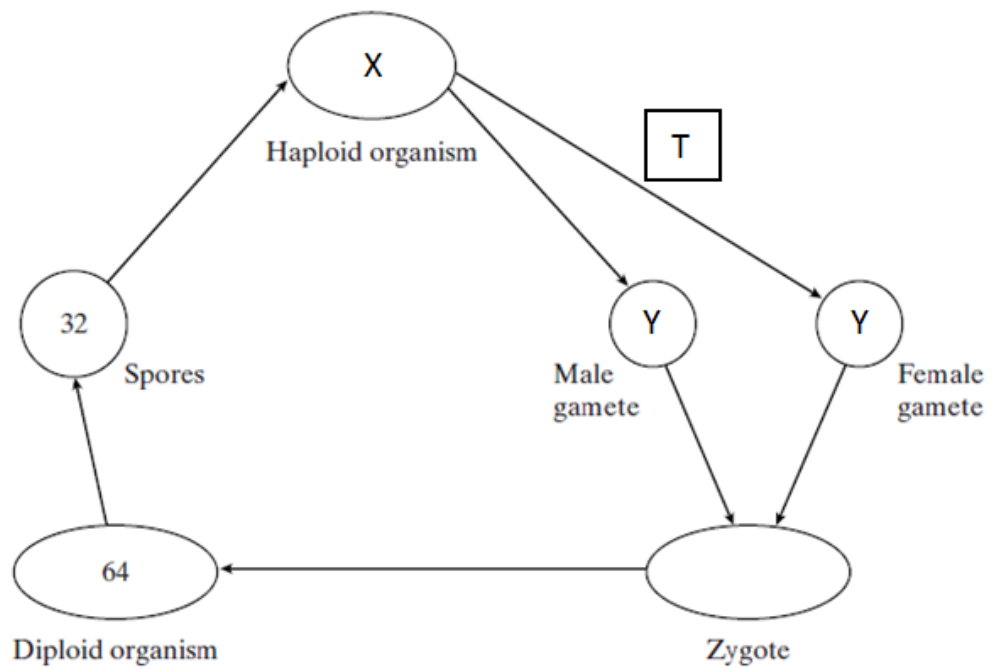
- 4 Curve **P** shows the rate of a reaction catalysed by lactate dehydrogenase under optimum conditions. A change was made to the reaction and curve **Q** shows the effect of this change on the reaction rate.



Which factor, operating to a constant extent throughout the experiment, could result in curve **Q**?

- A Addition of a compound that competes for the same binding site as pyruvate
 - B Addition of an inhibitor that differs in 3D configuration from pyruvate
 - C Addition of a co-enzyme such as NAD^+
 - D An increase in enzyme concentration
- 5 During the mitotic cell cycle, which of the following would result if cytokinesis does not occur after mitosis is completed?
- A Cells with two nuclei
 - B Cells with insufficient organelles
 - C Cells with twice the amount of genetic material but without nuclei
 - D Cells which are unusually small in size

- 6 The diagram shows the life cycle of an organism. The numbers show how many chromosomes are present in one cell at each stage of the life cycle.



Which of the following correctly shows the type of division and number of chromosomes?

	Type of cell division	Number of chromosomes	
	T	X	Y
A	Mitosis	16	8
B	Meiosis	16	16
C	Mitosis	32	32
D	Mitosis	32	16

- 7 The table shows the relative amounts of the bases adenine, thymine, guanine and cytosine in DNA from different organisms.

source	adenine	thymine	guanine	cytosine
bacterium	23.8	23.1	26.8	26.3
maize	26.8	27.2	22.8	23.2
fruit fly	30.7	29.5	19.6	20.2
chicken	28.0	28.4	22.0	21.6
human	29.3	30.0	20.7	20.0

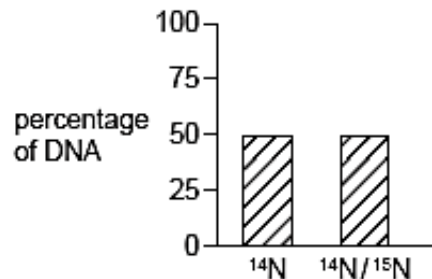
Which statements account for the importance of the ratios of A to T and G to C to the structure of DNA?

- 1 Complementary base pairing can occur.
- 2 Mutation will occur when pairing ratio is lost.
- 3 Semi-conservative DNA replication can occur to copy DNA strands.
- 4 Phosphodiester bonds helps to hold two strands together.
- 5 Purines and pyrimidines have different sizes and shapes.

- A** 1 and 3 only
B 1, 2, 3 and 5 only
C 2, 3, 4 and 5 only
D 1, 2, 3, 4 and 5

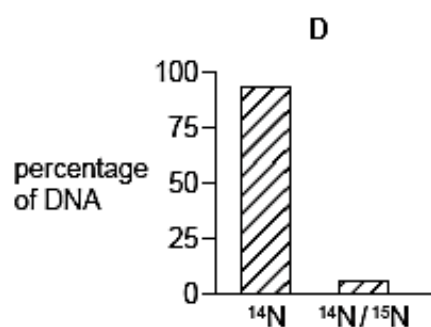
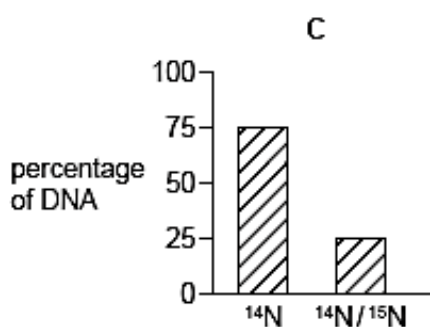
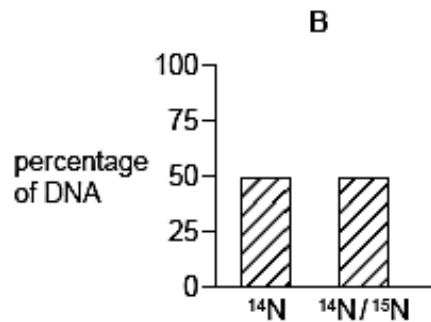
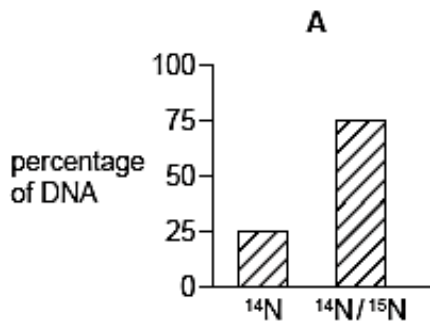
- 8 Bacteria were grown in a medium containing ^{15}N . After several generations, all of the DNA contained ^{15}N . Some of these bacteria were transferred to a medium containing the common isotope of nitrogen, ^{14}N . The bacteria were allowed to divide once. The DNA of some of these bacteria was extracted and analysed. This DNA was all hybrid DNA containing equal amounts of ^{14}N and ^{15}N .

Some bacteria from the medium with ^{15}N were transferred into a medium of ^{14}N . The bacteria were allowed to divide twice. The graph shows the percentages of ^{14}N and ^{15}N in the DNA of these bacteria.



Some bacteria from the medium with ^{15}N were transferred into a medium of ^{14}N . The bacteria were allowed to divide three times.

What would be the percentages of ^{14}N and ^{15}N in the DNA extracted from these bacteria?



- 9 The following statements illustrate the processes that occur during translation, although not necessarily in this order.
- 1 The large subunit of the ribosome binds and forms the translation initiation complex.
 - 2 The second amino acyl-tRNA complex now binds to mRNA at the “A” site of the ribosome.
 - 3 The small ribosomal subunit, with initiator tRNA bound, binds to the 5’ cap of the mRNA and scans for the first start codon.
 - 4 Soluble protein called release factor recognises the stop codon and binds at the “A” site.
 - 5 Formation of a peptide bond between the first and the second amino acids by peptidyl transferase.
 - 6 The second amino acyl-tRNA complex moves from the “A” site to the “P” site.

Using the information provided above, deduce the order in which these processes occur.

- A** 1 → 3 → 2 → 5 → 6 → 4
B 1 → 3 → 2 → 6 → 5 → 4
C 3 → 1 → 2 → 5 → 6 → 4
D 3 → 1 → 2 → 6 → 5 → 4

- 10 The following table shows the mRNA codons for six different amino acids.

mRNA codons	amino acid
AAA AAG	lysine
AGA AGG CGG	arginine
GGU GGA GGC GGG	glycine
CCU CCA CCC CCG	proline
UGG	tryptophan
UAU UAC	tyrosine

The base sequence of mRNA coding for part of a polypeptide is shown below.

U A U A A G A G G C C U U G G
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
 ↑
start reading

From the information provided, which of the predictions stated below is **not** true?

- A** The insertion of a nucleotide between positions 3 and 4 is expected to result in a greater change in the amino acid sequence than an insertion between positions 12 and 13.
- B** The deletion of a nucleotide at position 5 would result only in an alteration of the second amino acid in the chain.
- C** The substitution of a different nucleotide at position 12 would produce no alteration in the amino acid chain.
- D** The substitution of a different nucleotide at position 13 would result in the alteration of one amino acid.

11 Some statements about the phages are listed as follows:

- 1 All types of phages are capable of undergoing lytic and lysogenic cycles.
- 2 A phage usually undergoes lysogenic cycle because a larger number of progeny phages can be produced rapidly as compared to the lytic cycle.
- 3 The release of lysozyme upon rupturing of lysosome leads to the osmotic lysis of host bacterium.
- 4 The phage gene codes for a repressor protein that prevents the expression of prophage.

Which of these statements about the reproductive cycles of a phage are **not** true?

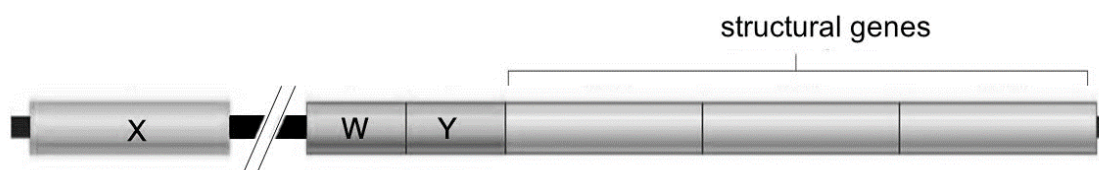
- A** 1 and 2 only
- B** 3 and 4 only
- C** 1, 2 and 3 only
- D** 2, 3 and 4 only

12 The dengue virus is spherical, membrane-bound with a similar reproduction cycle as the influenza virus. However, unlike the influenza virus, its genetic material consists of one single positive sense strand of RNA.

Using the information above, which one of the following statements about the reproduction cycle of dengue virus is **false**?

- A** The viral envelope fuses with the cell surface membrane of the host cell.
- B** The viral genome is directly used for translation of viral protein.
- C** RNA-dependent RNA polymerase is required to complete its cycle.
- D** Uncoating process involves fusion with endosome membrane.

- 13 The diagram shows a length of DNA responsible for metabolising lactose in prokaryotes. A mutation occurred in X such that its protein product became non-functional.



What is a possible outcome of the mutation if the bacteria cell is grown in a culture medium containing only lactose?

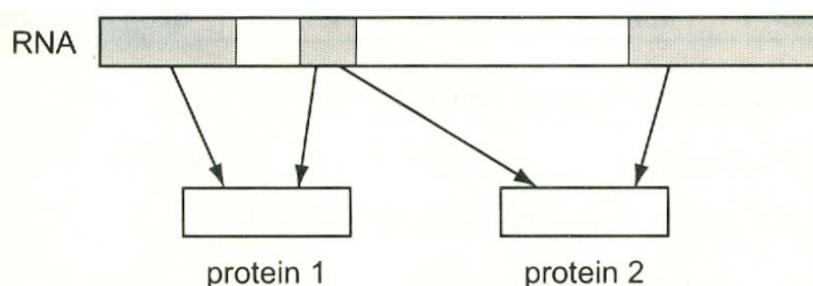
	Sequence bound by functional protein X	Positive control of gene regulation	Negative control of gene regulation	Transcription of structural genes
A	W	absent	present	no
B	Y	absent	present	no
C	W	present	absent	yes
D	Y	present	absent	yes

- 14 Which of the following statements is **true** about bacterial conjugation?
- A Cytoplasmic bridge started out as a sex pilus.
 - B DNA to be transferred by conjugation need to be fragmented before transfer.
 - C It requires two bacteria to be in direct contact with each other.
 - D Cytoplasmic bridge formation is dependent on the donor bacteria that have the competence factor.
- 15 About 12,000 genes are expressed in both chick liver and oviduct. However an estimated additional 5,000 genes are expressed only in liver, while an additional 3,000 genes are expressed only in oviduct.

Which of the following could explain these observations?

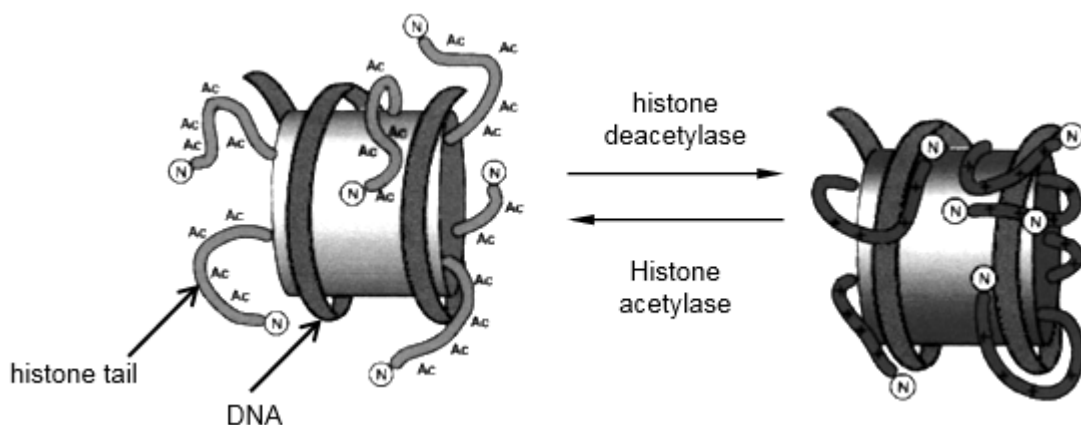
- 1 The additional genes may have different methylation patterns in different tissues.
 - 2 The concentrations of transcriptional enhancer elements for the additional genes vary in different tissues.
 - 3 The number of genome copies is different in different somatic cells.
 - 4 A common set of genes are expressed for normal functions in liver and oviduct.
- A 1 only
 - B 2 and 3 only
 - C 1 and 4 only
 - D 2 and 4 only

- 16 RNA transcribed from a length of DNA of a chromosome was found to code for two different proteins that function as enzymes, as shown in the diagram.



Which of the following best describes the two proteins?

- A Protein 1 and protein 2 are a result of control of gene expression at translational level.
 - B Protein 1 and protein 2 are a result of the cleavage of a polypeptide.
 - C Protein 1 and protein 2 function sequentially in the same metabolic pathway.
 - D Protein 1 and protein 2 usually perform similar functions in different cell types.
- 17 The diagram below shows how acetylation of histones promotes loose chromatin structure. Recent evidence has shown that chemical modification of histones play a direct role in regulation of gene expression.



Which of the following best explains how acetylation regulates gene expression?

- A Helicase action is enhanced by acetylation.
- B Acetylation of histones neutralizes their negative charges and encourages binding to DNA polymerase.
- C When nucleosomes are highly acetylated, chromatin becomes less compact and DNA is more accessible for transcription.
- D RNA polymerase works better by binding with acetyl groups.

- 18** Gene expression in eukaryotes can be regulated at the translational level.

Which combination of statements correctly describes eukaryotic translational control?

	Condition	Effect
A	Lack of translation initiation factor proteins	Inhibition of translation of selected mRNA
B	Presence of repressor proteins binding to 5'-UTR of selected mRNA	Prevents small ribosomal subunit from binding
C	Presence of repressor proteins to distal control elements	mRNA is translationally-repressed
D	mRNA with a long poly-A tail	mRNA is degraded slowly by restriction endonuclease

- 19** One type of genetic changes that can likely be found in the nucleus of a tumor cell is

- A** addition of nucleotides leading to frame-shift mutation in telomerase gene.
- B** tumor suppressor gene coming under the control of an active viral promoter.
- C** substitution mutation in proto-oncogene resulting in a premature stop codon.
- D** chromosomal deletion in region where tumor suppressor genes are located.

- 20 In a comparative study of brinjal plants, a test cross was made between the variety of plant producing purple and long brinjal and the variety producing green and short brinjal. The results of the following F_1 generation are shown below:

Phenotypes	Number
Purple, Long	28
Purple, Short	30
Green, Long	26
Green, Short	34
Total number	118

$$\chi^2 \text{ test} \quad \chi^2 = \sum \frac{(O - E)^2}{E} \quad \nu = c - 1$$

key to symbols

Σ = 'sum of ...'

ν = degrees of freedom

c = number of classes

O = observed 'value'

E = expected 'value'

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Which of the following statements is **false**?

- A The probability that the difference between observed and expected values is due to chance is greater than 10%.
- B The two genes coding for the colour and shape are not linked.
- C The difference between the expected number and the observed number of the phenotypes occurred by chance.
- D The calculated χ^2 value is greater than the critical χ^2 value.

- 21** An insect collection contains 102 specimens of a species of butterfly. This specimen is sexually dimorphic, meaning that the males and females look different from each other. A student examined the specimens and collected the following data.

<i>Observation</i>	<i>Frequency</i>
Blue wing colour (male)	64
Brown wing colour (female)	38
Wing span 35-37 mm	15
Wing span 37-39 mm	68
Wing span 39-41 mm	19

How should this variation be classified?

	continuous	discontinuous
A	sexual dimorphism	colour
B	colour	sexual dimorphism
C	sexual dimorphism	wingspan
D	wingspan	sexual dimorphism

- 22** In certain breeds of mice, the two allelic pairs, **C/c** and **A/a** are known to regulate the formation of coat colour. Coloured coat is produced in the breed carrying at least one copy of **C** allele and two copies of a alleles, whereas agouti coat is produced when the breed carries at least one copy of **C** and **A** alleles. Albino coat is derived from the breed homozygous for c allele.

Which of the following description is valid when two agouti mice heterozygous for both **C** and **A** genes are crossed?

- A** The expected ratio of agouti mice to coloured mice and to albino mice is 9:6:1.
- B** The phenotypic effect coded by the allele **A** is masked in the mice homozygous for allele **c**.
- C** The production of coat colour is regulated sequentially first by gene **A** and then gene **C**.
- D** Agouti is an intermediate coat colour resulting from the codominant effect between gene **A** and gene **C**.

- 23** In domestic poultry, the allele for white feathers (**W**) is dominant to the allele for dark feathers (**w**), and the allele for frizzled feathers (**F**) is dominant to the allele for normal feathers (**f**).

Poultry which were homozygous for dark, frizzled feathers were bred with poultry bearing white, normal feathers. The offspring of this cross were crossed with homozygous recessive individuals and the following results obtained:

<i>Phenotype</i>	<i>Number of offspring</i>
white, frizzled feather	18
dark, frizzled feather	63
white, normal feather	63
dark, normal feather	13

Which one of the following statement is consistent with these results?

- A** The genes are found on 2 separate pairs of homologous chromosomes.
 - B** The genes are linked on the same chromosome and allele **W** is linked to allele **w**.
 - C** The genes are linked on the same chromosome and allele **W** is linked to allele **F**.
 - D** The genes are linked on the same chromosome and allele **w** is linked to allele **F**.
- 24** In the breeding season, male Anole lizards court females by bobbing their heads up and down while displaying a colourful throat patch. Both characteristics are controlled by genes found on separate chromosomes. Anoles prefer to mate with lizards, which bob their heads fast and have red throat patches. These two alleles are dominant over their counterparts, slow bobbing and yellow throats.

A male lizard heterozygous for head bobbing and homozygous dominant for the red throat patch mates with a female that is also heterozygous for head bobbing but has yellow throat patch.

What percentage of the offspring has fast bobbing and red throat phenotype?

- A** 25%
- B** 50%
- C** 75%
- D** 100%

- 25** Two separate experiments were conducted to investigate the production of oxygen in photosynthesizing plants.

In experiment 1, an illuminated suspension of photosynthesizing algae *Chlorella* was given carbon dioxide containing a heavy isotope of oxygen, ^{18}O . The amount of radioactively labeled oxygen ($^{18}\text{O}_2$) produced from the *Chlorella* suspension was then measured.

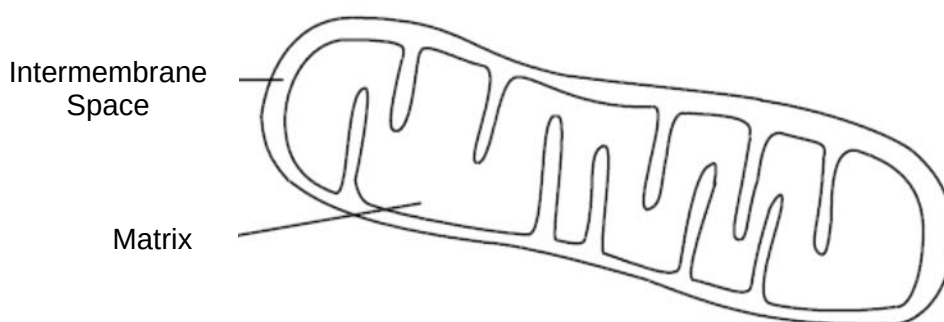
The experiment was then repeated but with *Chlorella* suspension given water molecules containing ^{18}O (experiment 2).

The results are shown in the table below.

Time after introducing ^{18}O / min	Amount of $^{18}\text{O}_2$ produced / arbitrary units	
	Experiment 1 (with C^{18}O_2)	Experiment 2 (with H_2^{18}O)
0	0	0
5	0	0.05
10	0	0.24
15	0	0.44
20	0	0.77

Which of the following conclusions can be inferred from the above experimental results?

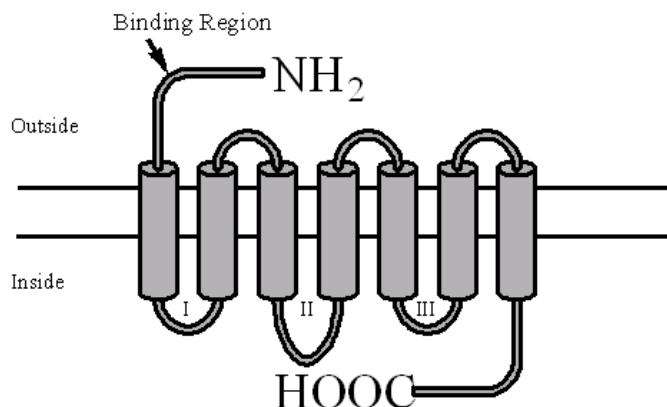
- A** Oxygen atoms in the carbon dioxide molecule is incorporated into Calvin cycle metabolites and subsequently sucrose molecules produced from photosynthesis.
 - B** Water is the source of oxygen evolved during photosynthesis.
 - C** Oxygen is the final electron acceptor of the non-cyclic photophosphorylation.
 - D** The rate of photosynthesis increases with time.
- 26** The diagram below shows a mitochondrion as seen under the electron microscope.



What is the advantage of having a small volume inside the inter-membrane space of the mitochondrion?

- A** A high electron concentration is rapidly developed.
- B** A high proton concentration is rapidly developed.
- C** Protein electron carriers are highly concentrated.
- D** ATP synthase is highly concentrated.

- 27 The figure below shows a protein receptor on a cell membrane.



Which statement correctly describes how this receptor responds upon binding to a ligand?

- A It enters the nucleus, undergoes a conformational change and binds to DNA.
- B It undergoes a conformational change, enters the nucleus and binds to DNA.
- C It undergoes conformational change and binds to G-protein.
- D It undergoes dimerization, and hence a conformational change.

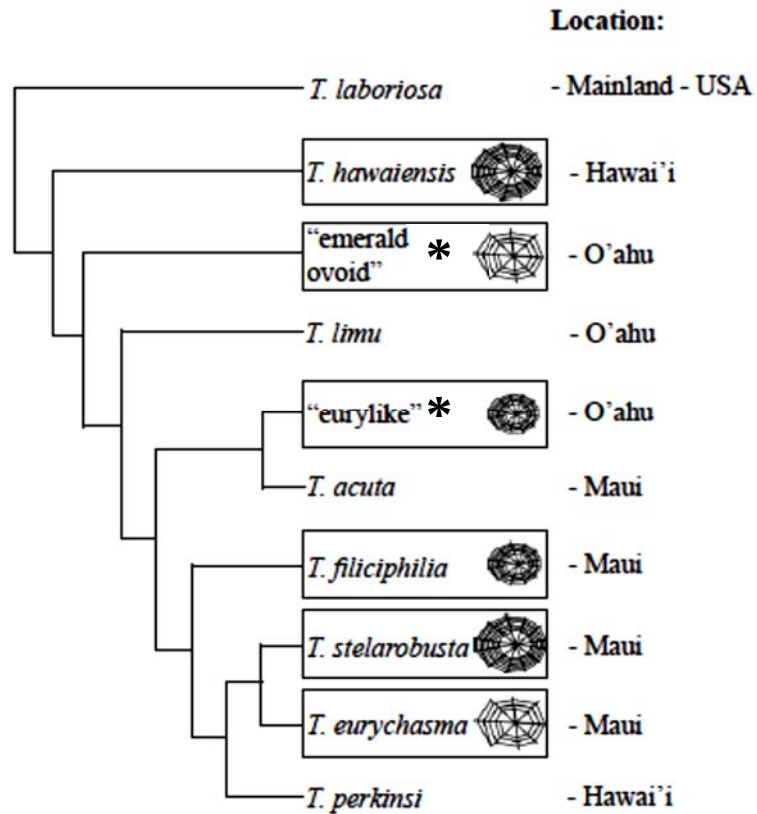
- 28 Which of the following statements correctly describes the effect of blood insulin?

	insulin level in blood	effect
A	high	a decrease in glucose metabolism
B	high	an increase in conversion of glycogen to glucose
C	low	decrease in glucose permeability of cells
D	low	an increase in uptake of glucose by muscle cells

- 29 During the propagation of the nerve impulse, the action potential results from the movement of

- A K^+ ions from intracellular fluid to extracellular fluid
- B K^+ ions from extracellular fluid to intracellular fluid
- C Na^+ ions from extracellular fluid to intracellular fluid
- D Na^+ ions from intracellular fluid to extracellular fluid

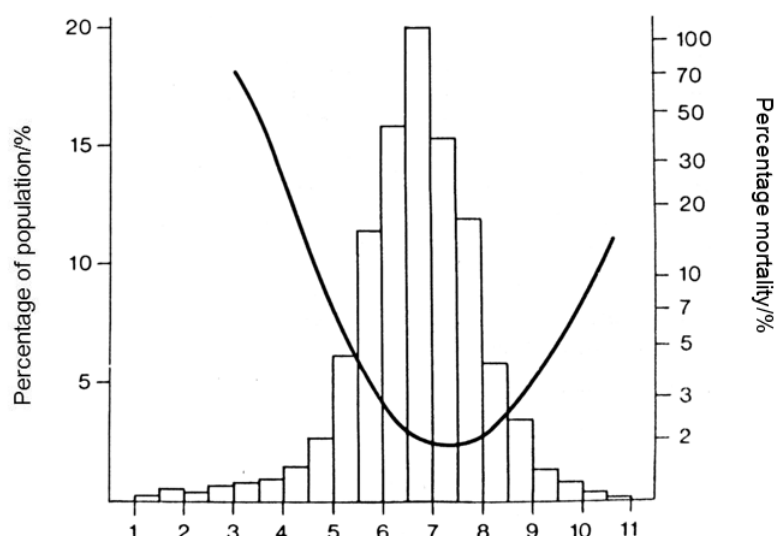
- 30 The cladogram below shows the classification of a group of spiders found on the Hawaiian islands. An asterisk (*) indicates that this species of spider has yet to be assigned a scientific name.



Based on information from the diagram above, deduce which of the following spiders are the most closely related species.

- A *T. filiciphilia* and "eurylike"
- B *T. hawaiiensis* and "emerald ovoid"
- C *T. stelarobusta* and *T. eurychasma*
- D *T. stelarobusta* and *T. filiciphilia*

- 31 The histogram represents the proportions of a population of new-born mammals falling into various birth weight classes. The line graph represents mortality.



From the information given, which conclusion is correct?

- A Birth weight is undergoing stabilising selection.
 B Birth weight is an example of discontinuous variation.
 C Birth weight is genetically linked to mortality.
 D Mortality is undergoing disruptive selection.
- 32 Calcitonin is a protein hormone found in humans, fish, birds, and mammals. It is 32 amino acids long. The table below shows the amino acid sequence of calcitonin in various organisms and the number of amino acid differences when compared with human calcitonin.

Organism	Amino acid sequence of calcitonin	Number of amino acid differences
Human	CGNLSTCMLGTYTQDFNKFHTFPQTAIGVGAP-NH ₂	-
Salmon	CSNLSTCVLGKLSQELHKLQTYPRTNTGSGTP-NH ₂	16
Eel	CSNLSTCVLGKLSQELHKLQTYPRTDVGAGTP-NH ₂	16
Rat	CGNLSTCMLGTYTQDLNKFHTFPQTSIGVGAP-NH ₂	2
Chicken	CASLSTCVLGKLSQELHKLQTYPRTDVGAGTP-NH ₂	17

What conclusion can be drawn from the data above?

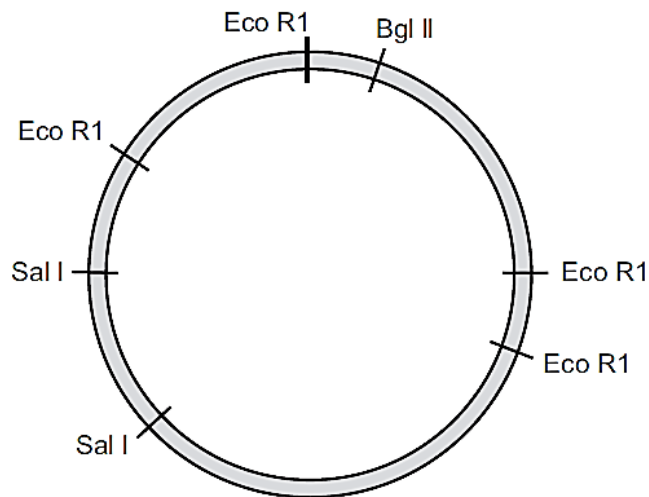
- A Humans and salmon are more closely related than salmon and eel.
 B Humans are most closely related to rats.
 C Salmon and chicken share a recent common ancestor.
 D Salmon and eel are more closely related than salmon and chicken.

33 Which of the following are basic requirements of Polymerase Chain Reaction?

- 1 heat stable DNA polymerase
- 2 two oligonucleotide primers
- 3 RNA segment to be amplified
- 4 free nucleotides

- A** 1, 2 and 3 only
B 1 and 2 only
C 1, 2 and 4 only
D 2 and 4 only

34 The following diagram indicates the cutting sites of three different restriction enzymes on a particular bacterial plasmid.



If the plasmid was completely digested with the restriction enzyme *Eco R1*, the number of pieces of DNA obtained would be

- A** 2
B 3
C 4
D 7

- 35 DNA fingerprints were obtained from four sources.

Crime scene sample	Victim	Suspect A	Suspect B
==	—		—
—		==	
—	—		
==		—	—
==			—
==	==	—	—

The sample taken from the crime scene contains DNA from the victim and

- A an unknown individual.
 - B Suspect A.
 - C Suspect B.
 - D Suspect A and Suspect B.
- 36 What is the correct sequence of events in Southern blotting?
- A Hybridization of DNA fragments with a labelled probe sequence followed by separation by electrophoresis and then transfer to a membrane.
 - B Separation of DNA fragments by electrophoresis followed by hybridization with a labelled probe sequence and then transfer to a membrane.
 - C Separation of DNA fragments by electrophoresis followed by transfer to a membrane and then hybridization with a labelled probe sequence.
 - D Transfer of DNA fragments to a membrane followed by separation by electrophoresis and then hybridization with a labelled probe sequence.
- 37 Cells from blastocyst offer great promise to the cure of many diseases because
- 1 the cells are pluripotent.
 - 2 the cells are more readily available than other cells.
 - 3 the cells will differentiate and produce cells which are normal.
 - 4 the cells can be easily cultured in vivo.
- A 2 only
 - B 3 only
 - C 1 and 3 only
 - D 1, 3 and 4 only

- 38 In terms of containment (prevention of genetic pollution of wild plant or non-GM crop populations), which of the following is an advantage of chloroplast transformation (introducing foreign genes into chloroplasts of host plants) over nuclear transformation?
- A Chloroplasts are surrounded by a double membrane.
 - B Chloroplasts are smaller than the nucleus.
 - C There are no chloroplasts in pollen of most plant species.
 - D There is no DNA in chloroplasts.
- 39 Which of the following is **not** an advantage of using tissue culture to cultivate transgenic plants?
- A Sexual reproduction is not necessary.
 - B Tissue culture reduces time taken for plant to grow to maturity.
 - C Plants are genetically identical and are susceptible to mass infections.
 - D Desirable characteristics can be passed on from one generation to the next.
- 40 A gene coding for a growth hormone is spliced together with an active promoter and introduced into Atlantic Salmon using genetic modification.

Which of the following facts about genetically modified (GM) Atlantic Salmon is **false**?

- A The GM salmon is larger and reaches sexual maturity sooner than wild salmon.
- B There is significant increase of growth hormone production in GM salmon.
- C The gene is obtained from Chinook salmon and the promoter sequence is obtained from Ocean pout.
- D The GM salmon may not be able to outcompete the wild salmon due to their large size.